



BSc (Hons) in **Computer Science**
SE3062 : Intelligent Systems

Year 3 · Semester 1 · 2026 Semester 2

Faculty of Computing

Practical 02

Objective & Lecture Mapping: In Lecture 03, we learned that a mathematically perfect "Table-Driven Agent" is impossible to build due to combinatorial explosion. Instead, we must write Agent Programs. Today, you will implement a **Simple Reflex Agent** using Condition-Action rules, observe its fatal flaws in a partially observable environment, and then upgrade it to a **Model-Based Agent** that maintains an internal memory state.

Part 1: Practical Implementation — Coding the Architectures

Step 1.1: Setting the Trap (Partial Observability)

To see why Simple Reflex agents fail, we must handicap your agent's sensors. We are moving from a Fully Observable world to a Partially Observable one.

1. Open your `visual_grid_game.py` from Lab 01.
2. Locate the `get_percept(self)` method.
3. Modify it so it **no longer returns** the exact global coordinates (`agent_pos`). Instead, it should only return local booleans, e.g., `{'wall_ahead': True/False, 'food_here': True/False}` based on checking the adjacent cell in its current facing direction.

Step 1.2: The Simple Reflex Agent (Implementation & Failure)

1. Create a new class called `SimpleReflexAgent` .
2. Implement a `sense_and_act(self, percept)` method that uses strictly IF-THEN logic (Condition-Action rules). *Do not use an `__init__` method to store history.*
Example Logic: IF `food_here` THEN `suck`; IF `wall_ahead` THEN `turn_left`; ELSE `move_forward`
3. Run the simulation. **Observe:** Watch the agent get trapped in a corner or a U-shaped wall, infinitely repeating a cycle (e.g., turn left, step forward, hit wall, turn left...).

Step 1.3: The Model-Based Agent (Memory & State)

1. Create a new class called `ModelBasedAgent` .
2. Implement an `__init__(self)` method to initialize an internal memory state (e.g., `self.visited_cells = set()` or a relative movement tracker).
3. In `sense_and_act(self, percept)` , first **update the state** (Transition & Sensor Model) by recording the current percept and the last action taken.
4. Update your IF-THEN rules to query the memory.
Example Logic: IF `wall_ahead` AND `left_is_visited` THEN `turn_right`
5. Run the simulation. **Observe:** The agent should now remember where it has been, realize it is in a loop, and choose an alternate path to escape.

Part 2: Theoretical Evaluation & Lecture Mapping

Based on your practical observations above, answer the following questions to map your code directly to the architectural theories covered in Lecture 02.

1. **(Remember) According to Lecture 02, why is it impossible to program a mathematically perfect "Table-Driven Agent" for complex environments like Chess? What happens as the agent's lifetime increases?**

A mathematically perfect Table-Driven Agent is impractical for complex environments such as Chess because it would need to store an action for every possible percept sequence. The number of possible percept sequences increases exponentially as the environment becomes more complex and as the agent's

2. (Understand) Look at the code you wrote for your

`SimpleReflexAgent`. Identify and explain the specific lines of code that represent the "Condition-Action Rules" discussed in the lecture.

forward.

These rules demonstrate the main characteristic of a Simple Reflex Agent because the agent selects an action based only on the current percept and does not use a stored percept history.

3. (Analyze) Your `SimpleReflexAgent` likely got stuck in an infinite loop during Step 1.2. Based on the lecture, analyze exactly why this happened. How did the combination of "Partial Observability" and a lack of "Percept History" cause this failure?

such as whether there is a wall ahead, whether food is present, or whether there is a wall on another side. It does not have complete information about the entire environment or its exact previous path.

The Simple Reflex Agent also does not remember previous actions or locations.

4. (Evaluate) In Step 1.3, you added an internal state to your `ModelBasedAgent`. Evaluate how your specific code handles the "Transition Model" (how the world evolves) and the "Sensor Model" (how the agent's actions affect the world).

the agent to remember locations that it has already visited.

This internal state allows the Model-Based Agent to make decisions using both current percepts and information from previous experiences. Therefore, when the agent detects a situation that has occurred before, it can use its stored state to

Finalize and Lock Lab 02 Documentation



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